

# NEPAL COLLEGE OF INFORMATION TECHNOLOGY

Balkumari, Lalitpur

*Affiliated to Pokhara University*

## ASSIGNMENT FOR COMPUTER GRAPHICS



## ASSIGNMENT 3

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Assignment 3

- 1) A point  $P(20, 20)$  in a circular window with its center at  $(100, 100)$  and a radius of 90 pixels is required to be mapped to a point in circular viewport having its at  $(50, 50)$  and radius of 40 pixels. Where will the point  $P$  be placed after the transformation?

Soln

Radius <sup>in</sup> window ( $w_r$ ) = 90

$(x_{wc}, y_{wc}) = (100, 100)$

Radius <sup>in</sup> viewport ( $v_r$ ) = 40

$(x_{vc}, y_{vc}) = (50, 50)$

$$S_x = S_y = \frac{v_r}{w_r} = \frac{40}{90} = 0.44$$

$$S_y = \frac{v_r}{w_r} = 0.44$$

$$\text{Composite Matrix (Cm)} = T(x_{vc}, y_{vc}) \cdot S_{uv} \cdot T(-x_{wc}, -y_{wc})$$

$$= \begin{bmatrix} 1 & 0 & 50 \\ 0 & 1 & 50 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 0.44 & 0 & 0 \\ 0 & 0.44 & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 & -100 \\ 0 & 1 & -100 \\ 0 & 0 & 1 \end{bmatrix}$$

$$= \begin{bmatrix} 0.44 & 0 & 6 \\ 0 & 0.44 & 6 \\ 0 & 0 & 1 \end{bmatrix}$$

$$\text{So, } P' = \text{Cm} \cdot P$$

$$= \begin{bmatrix} 0.44 & 0 & 6 \\ 0 & 0.44 & 6 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 20 \\ 20 \\ 1 \end{bmatrix} = \begin{bmatrix} 14.8 \\ 14.8 \\ 1 \end{bmatrix}$$



Hence, the point  $P(20, 20)$  after being mapped to viewport will be  $P'(14.8, 14.8) \approx P'(15, 15)$ . Ans

Q2) A triangle with vertices  $A(5, 2)$ ,  $B(4, 1)$ ,  $C(6, 1)$  is required to be rotated in a clockwise direction by  $45$  degrees about any arbitrary point  $(4, 4)$ . Find out the final coordinate positions of the triangle after performing the desired transformation.

Sol

Here,  $\theta = -45^\circ$  (cw)

Now,

$$\text{composite matrix } (C_m) = T(4, 4) \cdot R_{\theta_{-45^\circ}} \cdot T(-4, -4)$$

$$= \begin{bmatrix} 1 & 0 & 4 \\ 0 & 1 & 4 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} \cos(-45) & -\sin(-45) & 0 \\ \sin(-45) & \cos(-45) & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 & -4 \\ 0 & 1 & -4 \\ 0 & 0 & 1 \end{bmatrix}$$

$$= \begin{bmatrix} 1 & 0 & 4 \\ 0 & 1 & 4 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} \cos 45 & \sin 45 & 0 \\ -\sin 45 & \cos 45 & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 & -4 \\ 0 & 1 & -4 \\ 0 & 0 & 1 \end{bmatrix}$$

$$= \begin{bmatrix} 0.707 & 0.707 & -1.66 \\ -0.707 & 0.707 & 4 \\ 0 & 0 & 1 \end{bmatrix}$$

Hence,  $P' = C_m \cdot P$

$$= \begin{bmatrix} 0.707 & 0.707 & -1.66 \\ -0.707 & 0.707 & 4 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 5 & 4 & 6 \\ 2 & 1 & 1 \\ 1 & 1 & 1 \end{bmatrix}$$





$$= \begin{bmatrix} 3.29 & 1.87 & 3.29 \\ 1.87 & 1.87 & 0.46 \\ 1 & 1 & 1 \end{bmatrix}$$

$$\approx \begin{bmatrix} 3 & 2 & 3 \\ 2 & 2 & 0 \\ 1 & 1 & 1 \end{bmatrix}$$

Hence, the vertices of triangle after transformation are  
 $A'(3, 2), B'(2, 2), C'(3, 0)$ .

Q3) Reflect a triangle  $A(1, 0), B(3, 1), C(1, 2)$  about line  $y = -2x + 5$  then scale it about a fixed point  $P(10, 10)$ .

Soln

Here,  $y = -2x + 5$

So,  $m = -2$  ,  $c = 5$

or,  $\theta = \tan^{-1}(-2)$

$\therefore \theta = -63.43^\circ$

lets assume scaling factor as '2' in both directions.

$$C_m = S_{m=2, x=2} \cdot T_{(0,5)} \cdot R_{\theta_{cw}} \cdot R_{\theta_{ccw}} \cdot R_{\theta_{cw}} \cdot T_{(0,-5)}$$

$$= \begin{pmatrix} 2 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 5 \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} \cos 63.43^\circ & \sin 63.43^\circ & 0 \\ -\sin 63.43^\circ & \cos 63.43^\circ & 0 \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} 1 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} \cos 63.43^\circ & -\sin 63.43^\circ & 0 \\ \sin 63.43^\circ & \cos 63.43^\circ & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

$$\begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & -5 \\ 0 & 0 & 1 \end{pmatrix}$$



$$\begin{aligned}
 &= \begin{pmatrix} 2 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 5 \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} -0.599 & -0.80 & 4 \\ -0.80 & 0.599 & -2.99 \\ 0 & 0 & 1 \end{pmatrix} \\
 &= \begin{pmatrix} -1.19 & -1.60 & 8.00 \\ -1.60 & 1.19 & 4.00 \\ 0 & 0 & 1 \end{pmatrix}
 \end{aligned}$$

So,  $P' = C_m \cdot P$

$$\begin{aligned}
 &= \begin{pmatrix} -1.19 & -1.60 & 8 \\ -1.60 & 1.19 & 4 \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} 1 & 3 & 1 \\ 0 & 1 & 2 \\ 1 & 1 & 1 \end{pmatrix} \\
 &= \begin{pmatrix} 6.81 & 2.83 & 3.61 \\ 2.4 & 0.39 & 4.78 \\ 1 & 1 & 1 \end{pmatrix}
 \end{aligned}$$

Hence, Final coordinates are

$(7, 2), (3, 0), (4.4, 5)$  Ans

Q4) A triangle with vertices  $A(5, 2), B(4, 1), C(6, 1)$  is required to be reflected about an arbitrary line  $y = 2x + 1$ . Find out the final coordinate positions of the triangle after performing the desired transformation.

Soln

Given line,  $y = 2x + 1$

So,  $m = 2$

or,  $\theta = \tan^{-1}(2)$

$\therefore \theta = 63.43^\circ$

Composite Matrix ( $C_m$ ) =  $R_{\theta}^{63.43} R_{\text{refl}} R_{\theta}^{-63.43}$

$$\begin{aligned}
 &= \begin{bmatrix} \cos 63.43^\circ & -\sin 63.43^\circ & 0 \\ \sin 63.43^\circ & \cos 63.43^\circ & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} \cos \theta & \sin \theta & 0 \\ \sin \theta & \cos \theta & 0 \\ 0 & 0 & 1 \end{bmatrix}
 \end{aligned}$$



$$= \begin{bmatrix} -0.59 & 0.81 & 0 \\ 0.81 & 0.59 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

Now,  $P' = C_m \times P$

$$= \begin{bmatrix} -0.59 & 0.81 & 0 \\ 0.81 & 0.59 & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 5 & 4 & 6 \\ 2 & 1 & 1 \\ 1 & 1 & 1 \end{bmatrix}$$

$$= \begin{bmatrix} -1.33 & -1.55 & -2.73 \\ 5.23 & 3.83 & 5.45 \\ 1 & 1 & 1 \end{bmatrix}$$

$$\approx \begin{bmatrix} -1 & -2 & -3 \\ 5 & 4 & 5 \\ 1 & 1 & 1 \end{bmatrix}$$

Hence, vertices after transformation are  $A'(-1, 5)$ ,  $B'(-2, 4)$ ,  $C'(-3, 5)$ .

Q5) A triangle with vertices  $A(5, 2)$ ,  $B(4, 1)$ ,  $C(6, 1)$  is required to be rotated by 45 degrees in CC direction about (i) origin and (ii) line  $y = 5$ .

Soln

i) origin

Step 1: Rotate the object by  $45^\circ$  in CCW.

$$R_{\theta=45^\circ, \text{CCW}} = \begin{bmatrix} \cos 45 & -\sin 45 & 0 \\ \sin 45 & \cos 45 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

The required component matrix is as,



$$= \begin{bmatrix} 0.7071 & -0.7071 & 0 \\ 0.7071 & 0.7071 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$\text{Now, } \begin{bmatrix} 0.7071 & -0.7071 & 0 \\ 0.7071 & 0.7071 & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 5 & 4 & 6 \\ 2 & 1 & 1 \\ 1 & 1 & 1 \end{bmatrix}$$

$$= \begin{bmatrix} 2.1213 & 2.1213 & 3.5355 \\ 4.9497 & 3.5355 & 4.9497 \\ 1 & 1 & 1 \end{bmatrix}$$

(ii)  $y=5$  line :

Step 1 : Translate the line  $y=5$  to  $x$ -axis.

Step 2 : Rotate the object by  $45^\circ$  in ccw.

Step 3 : Translate back the line along with rotated object to original position

The composite matrix is as,

$$C_m = T(0,5) \cdot R_{45^\circ}^{ccw} \cdot T(0,-5)$$

$$C_m = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 5 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} \cos 45 & -\sin 45 & 0 \\ \sin 45 & \cos 45 & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & -5 \\ 0 & 0 & 1 \end{bmatrix}$$

$$= \begin{bmatrix} 0.7071 & -0.7071 & 0 \\ 0.7071 & 0.7071 & 5 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & -5 \\ 0 & 0 & 1 \end{bmatrix}$$

$$= \begin{bmatrix} 0.7071 & -0.7071 & 3.53 \\ 0.7071 & 0.7071 & 1.46 \\ 0 & 0 & 1 \end{bmatrix}$$



Now, The matrix coordinates after rotation is,

$$P' = \begin{bmatrix} 0.7071 & -0.7071 & 3.53 \\ 0.7071 & 0.7071 & 1.46 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 5 & 4 & 6 \\ 2 & 1 & 1 \\ 1 & 1 & 1 \end{bmatrix}$$

$$= \begin{bmatrix} 5.6513 & 5.6513 & 7.0655 \\ 6.4097 & 4.9955 & 6.4097 \\ 1 & 1 & 1 \end{bmatrix}$$

Hence, the points are  $A'(6.6)$ ,  $B'(6.5)$ ,  $C'(7.6)$ .

Q6) Clip a line with end point coordinates  $A(-1, 6)$ ,  $B(5, -8)$  against a clip window with its lower left corner at  $(-2, -5)$  and upper right corner at  $(4, 8)$  using cohen-sutherland algorithm.

Soln

step 1: Establish region codes for line end points.

| $A(-1, 6)$ |           |   |     | $B(5, -8)$ |           |   |   |
|------------|-----------|---|-----|------------|-----------|---|---|
| L:         | $-1 < -2$ | F | '0' | L:         | $5 < -2$  | F | 0 |
| R:         | $-1 > 4$  | F | '0' | R:         | $5 > 4$   | T | 1 |
| B:         | $6 < -5$  | F | '0' | B:         | $-8 < -5$ | T | 1 |
| T:         | $6 > 8$   | F | '0' | T:         | $-8 > 8$  | F | 0 |

So, Bits = 0000

Bits for B = 0110

So, this is not the case of full visibility.



Step 2: Perform AND operation of region codes.

$$\begin{array}{r}
 0000 \\
 0110 \\
 \hline
 0000
 \end{array}$$

So, this is the case of partial visibility.

Ex/Ex.

Here, slope (m) =  $\frac{y_2 - y_1}{x_2 - x_1} = \frac{-8 - 6}{5 - (-1)} = \frac{-14}{6}$

Step 3: Determine intersection points of line with window boundaries.

For B(5, -8),

For bit 2,

$y = 8$        $x = x_{\text{window}} = 4$   
 and,  ~~$x$~~

and,  $y = y_1 + m(x - x_1)$   
 $= -8 + \frac{-14}{6}(4 - (-1))$

$\therefore y = \frac{-31}{3} = -10.33 \approx -10 \therefore y = \frac{-17}{3}$

$(x', y') = (4, \frac{-17}{3})$        $(x', y') = (4, \frac{-17}{3})$

For bit 3,  $y = y_{\text{window}} = -5$

and,

$x = x_1 + \frac{1}{m}(y - y_1)$

$= 4 + \frac{6}{-14}(-5 - (\frac{-17}{3}))$



$$B_x m = 1.43 \times 3.71$$

$$\therefore x \approx 4$$

$$\text{Hence, } B \text{ is } (x'', y'') = (4, -5)$$

The coordinates of the end points of visible line segment after clipping are  $A'(-1, 6)$  and  $B'(4, -5)$ . Ans

Q7) Derive the composite transformation matrix that reflects an object about line 'L' with necessary figures.

Soln

Let the line 'L' be considered as  $y = mx + c$ .

Step 1: The line is translated through factor  $(0, -c)$  along with the object to be reflected, such that the line passes through origin.

Step 2: Now, the line along with the object is rotated until it aligns with one of the major coordinate axes (consider X-axis) by angle  $\theta$ .

$$\text{where, } \theta = \tan^{-1}(m) \quad \xrightarrow{\text{value}} \text{Derived from eq?}$$

Step 3: The reflection of the object is taken along x-axis.

Step 4: After reflection, the line along with the object is rotated back to the same slope with angle ' $\theta$ ' but in opposite direction.

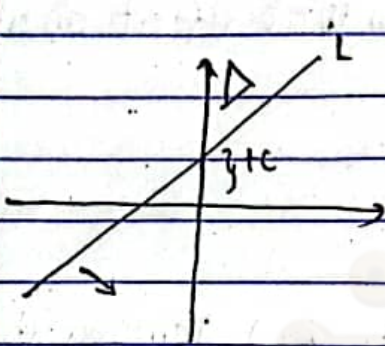
Step 5: Now, we perform inverse translation to bring the line back to its initial location along with the reflected object.



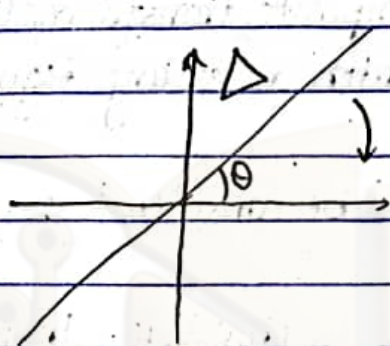
The composite matrix is,

$$C_m = T_{(0,c)} \cdot R_{ocw} R_{fx} \cdot R_{ocw} \cdot T_{(0,-c)}$$

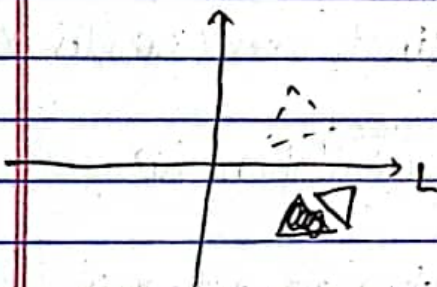
$$= \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & c \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} \cos\theta & -\sin\theta & 0 \\ +\sin\theta & \cos\theta & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} \cos\theta & \sin\theta & 0 \\ -\sin\theta & \cos\theta & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & -c \\ 0 & 0 & 1 \end{bmatrix}$$



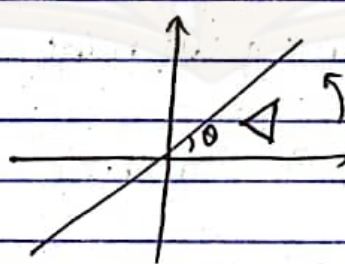
Step 1: Translation



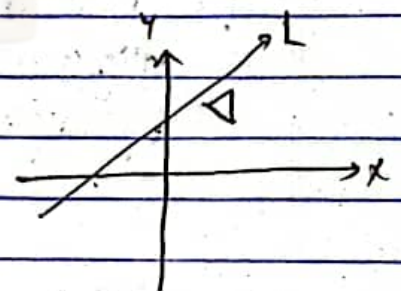
Step 2: Rotation



Step 3: Reflection



Step 4: Inverse Rotation



Step 5: Inverse translation

Q8) Find scaling transformation matrix to scale  $s_x, s_y$  units with respect to fixed point  $P(x, y)$ .

soln



Step 1: The fixed point  $P(x, y)$  along with the object to be rotated about it, is translated to origin.

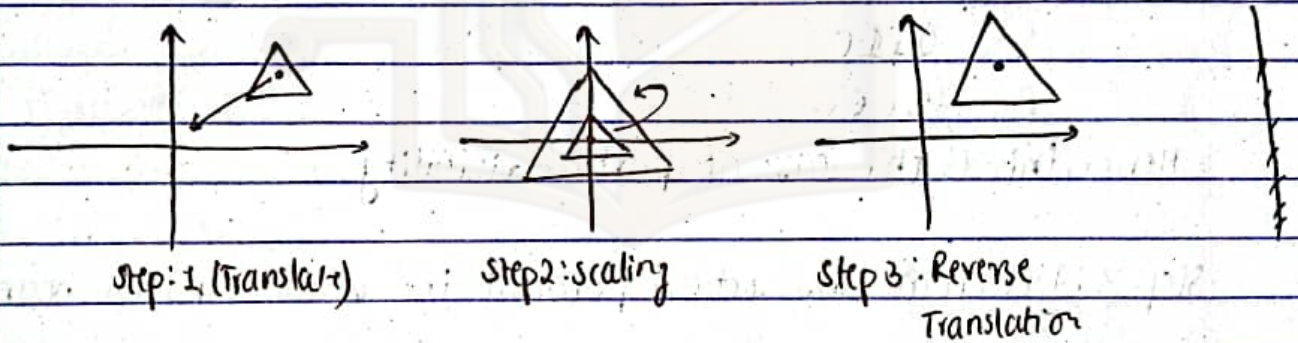
Step 2: Now scale the object about origin by the desired scaling factor.

Step 3: Translate back the fixed point which is at the coordinate origin along with the scaled object to its original location.

The composite matrix for these transformations is as follows:

$$C_m = T(x, y) \cdot S_{s_x, s_y} \cdot T^{-1}(x, y)$$

$$= \begin{pmatrix} 1 & 0 & x \\ 0 & 1 & y \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} s_x & 0 & 0 \\ 0 & s_y & 0 \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} 1 & 0 & -x \\ 0 & 1 & -y \\ 0 & 0 & 1 \end{pmatrix}$$

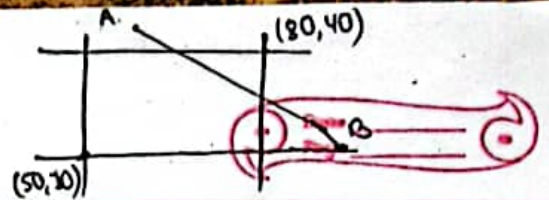


Q9) Use Cohen Sutherland's algorithm to clip line  $(60, 50) (100, 10)$  against window  $(50, 10) (80, 40)$ .

soln



Given points, A(60,50) B(100,10)



$$(x_{wmin}, y_{wmin}) = (50, 10)$$

$$(x_{wmax}, y_{wmax}) = (80, 40)$$

Step 1: Establish region codes for line end points.

A(60,50):

$$L: 60 < 50 \quad F \quad 0$$

$$R: 60 > 80 \quad F \quad 0$$

$$B: 50 < 10 \quad F \quad 0$$

$$T: 50 > 40 \quad T \quad 1$$

$$\text{Region code} \Rightarrow 0001$$

B(100,10)

$$L: 100 < 50 \quad F \quad 0$$

$$R: 100 > 80 \quad T \quad 1$$

$$B: 10 < 10 \quad F \quad 0$$

$$T: 10 > 40 \quad F \quad 0$$

$$\text{Region code} \Rightarrow 0010$$

Hence, this isn't case of full visibility.

Step 2: Perform logical AND operation of region codes.

$$0001$$

$$0010$$

$$0000$$

Hence, this is the case of partial visibility.

Step 3: Determine intersection point of line with window boundaries.

For A,  $(x_1, y_1) = (60, 50)$

$$y = y_{wmax} = 40$$

$$x = \frac{1}{m} (y - y_1) + x_1$$

$$= \frac{1}{-1} (40 - 50) + 60$$

$$\therefore x = 70$$

$$m = \frac{y_2 - y_1}{x_2 - x_1}$$

$$= \frac{10 - 50}{100 - 60}$$

$$= \frac{-40}{40}$$

$$= -1$$



For B,  $(x_1, y_1) = (100, 10)$

$$x = x_{\text{wmax}} = 80$$

$$y = m(x - x_1) + y_1$$

$$= -1(80 - 100) + 10$$

$$\therefore y = 30$$

Hence, the coordinates of the end points of visible line segment after clipping are ~~A(100, 10)~~, A'(70, 40) and B'(80, 30). Ans

Q10) show that 2D reflection through  $x$ -axis followed by 2D reflection through line  $y = -x$  is equivalent to a pure rotation (90 degrees) about origin.

Sol<sup>n</sup>

To prove, let's deduce the composite matrix  $C_{m1}$  and  $C_{m2}$  for two cases.

For first case,

$C_{m1} =$

$R_{f/n}$  Given line,  $y = -x + 0$

$$\text{So, } m = -1$$

$$\therefore \theta = -45^\circ$$

$$\text{Composite matrix for line reflection} = R_{\theta_{\text{ccw}}}^{-45^\circ} \cdot R_{f/n} \cdot R_{\theta_{\text{ccw}}}^{45^\circ}$$

Then, Final  $C_m$  with reflection through  $x$ -axis is,

$$C_{m1} = R_{\theta_{\text{ccw}}}^{-45^\circ} \cdot R_{f/n} \cdot R_{\theta_{\text{ccw}}}^{45^\circ} \cdot R_{f/x}$$

$$= \begin{pmatrix} \cos 45^\circ & \sin 45^\circ & 0 \\ -\sin 45^\circ & \cos 45^\circ & 0 \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} 1 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} \cos 45^\circ & -\sin 45^\circ & 0 \\ \sin 45^\circ & \cos 45^\circ & 0 \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} 1 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$$



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$$\text{So, } C_{m1} = \begin{pmatrix} 0 & 1 & 0 \\ -1 & 0 & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

Now, For the second case,

$$C_{m2} = R_{\theta_{cw}^{-90^\circ}}$$

$$= \begin{pmatrix} \cos 90^\circ & \sin 90^\circ & 0 \\ -\sin 90^\circ & \cos 90^\circ & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

$$= \begin{pmatrix} 0 & 1 & 0 \\ -1 & 0 & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

Hence, we can observe  $C_{m1} = C_{m2}$

So, 2D reflection through  $x$ -axis followed by 2D reflection through line  $y = -x$  is equivalent to a pure rotation ( $90^\circ$ ) about origin.

Q11) soln Prove that scaling followed by rotation is equivalent to shearing.



Q12) Triangle with vertices  $A(1,1)$ ,  $B(7,1)$ ,  $C(4,3)$  is required to be rotated about any arbitrary fixed point  $(4,2)$  in a counter clock wise direction by  $90$  degrees. What will be the final coordinates of the triangle?

Soln

$$\text{Composite Matrix } (C_m) = T_{(4,2)} R_{0^{90^\circ} \text{ ccw}} T_{(-4,-2)}$$

$$= \begin{pmatrix} 1 & 0 & 4 \\ 0 & 1 & 2 \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} \cos 90^\circ & -\sin 90^\circ & 0 \\ \sin 90^\circ & \cos 90^\circ & 0 \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} 1 & 0 & -4 \\ 0 & 1 & -2 \\ 0 & 0 & 1 \end{pmatrix}$$

$$= \begin{pmatrix} 0 & -1 & 8 \\ 1 & 0 & -2 \\ 0 & 0 & 1 \end{pmatrix}$$

So, Final coordinates,  $P' = C_m \cdot P$

$$= \begin{pmatrix} 0 & -1 & 8 \\ 1 & 0 & -2 \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} 1 & 7 & 4 \\ 1 & 1 & 3 \\ 1 & 1 & 1 \end{pmatrix}$$

$$= \begin{pmatrix} 5 & 5 & 3 \\ -1 & 5 & 2 \\ 1 & 1 & 1 \end{pmatrix}$$

Hence, final coordinates triangle after rotation are

$A(5,-1)$ ,  $B(5,5)$ ,  $C(3,2)$ . Ans

Q13) A point  $P(20,20)$  in a window with its lower most left corner at  $(10,10)$  and uppermost right corner at  $(100,100)$  is required to be mapped to a point in viewport having its lower most left corner at  $(30,30)$  and uppermost right corner at  $(90,80)$ . Where will





the point, P be placed after the transformation?

Soln

Given,  $P(20, 20)$

$$(x_{wmin}, y_{wmin}) = (10, 10)$$

$$(x_{wmax}, y_{wmax}) = (100, 100)$$

$$(x_{vmin}, y_{vmin}) = (30, 30)$$

$$(x_{vmax}, y_{vmax}) = (90, 80)$$

For this,

$$\text{Composite matrix} = T_{(x_{vmin}, y_{vmin})} \cdot S_{sx, sy} \cdot T_{(x_{wmin}, y_{wmin})} \quad \text{--- (1)}$$

$$\text{Here, } S_x = \frac{x_{vmax} - x_{vmin}}{x_{wmax} - x_{wmin}}$$

$$= \frac{90 - 30}{100 - 10} = \frac{60}{90} = 0.67$$

$$S_y = \frac{y_{vmax} - y_{vmin}}{y_{wmax} - y_{wmin}}$$

$$= \frac{80 - 30}{100 - 10} = \frac{50}{90} = 0.56$$

$$\begin{aligned} \text{So, } C_m &= \begin{pmatrix} 1 & 0 & 30 \\ 0 & 1 & 30 \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} 0.67 & 0 & 6 \\ 0 & 0.56 & 0 \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} 1 & 0 & -10 \\ 0 & 1 & -10 \\ 0 & 0 & 1 \end{pmatrix} \\ &= \begin{pmatrix} 0.67 & 0 & 23.3 \\ 0 & 0.56 & 24.4 \\ 0 & 0 & 1 \end{pmatrix} \end{aligned}$$

$$\text{So, } P' = C_m \cdot P$$

$$= \begin{pmatrix} 0.67 & 0 & 23.3 \\ 0 & 0.56 & 24.4 \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} 20 \\ 20 \\ 1 \end{pmatrix} = \begin{pmatrix} 36.7 \\ 35.6 \\ 1 \end{pmatrix} \approx \begin{pmatrix} 37 \\ 36 \\ 1 \end{pmatrix}$$

Hence, the point P will be placed at  $(37, 36)$  Ans



Q14) Derive a composite transformation matrix for mapping a point  $P_{\text{for}}$  in a circular window with a radius of 50 pixels and having its center at (50, 50) to a circular view port with a radius of 20 pixels and having its center at (10, 10).

Soln

Here,  $(x_{wc}, y_{wc}) = (50, 50)$

$(x_{vc}, y_{vc}) = (10, 10)$

Then,  $w_r = 50$  pixels  $v_r = 20$  pixels

So,  $s_x = s_y = \frac{v_r}{w_r} = \frac{20}{50} = 0.4$

$C_m = T(x_{vc}, y_{vc}) \cdot S_{s_x, s_y} \cdot T(-x_{wc}, -y_{wc})$

$$= \begin{pmatrix} 1 & 0 & 10 \\ 0 & 1 & 10 \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} 0.4 & 0 & 0 \\ 0 & 0.4 & 0 \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} 1 & 0 & -50 \\ 0 & 1 & -50 \\ 0 & 0 & 1 \end{pmatrix}$$

$$= \begin{pmatrix} 0.4 & 0 & -10 \\ 0 & 0.4 & -10 \\ 0 & 0 & 1 \end{pmatrix}$$

Q15) A window has its lower most left corner at (-5, 10) and its upper most right corner at (35, 40). clip a line segment that has two end points (-20, 5) and (60, 20). If a viewport is defined by (6, 17) and (90, 80). Find the view port coordinates for the intersection points.

Soln



$$(x_{wmin}, y_{wmin}) = (-5, 10)$$

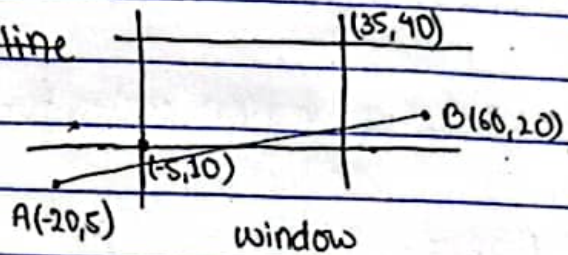
$$(x_{wmax}, y_{wmax}) = (35, 40)$$

$$(x_{vmin}, y_{vmin}) = (6, 17)$$

$$(x_{vmax}, y_{vmax}) = (90, 80)$$

End points of line : A (-20, 5) and B (60, 20)

Step 1: Establish region codes for line



First, let's translate transform points to viewport.

$$\begin{aligned} \text{For } S_x, S_y &= \frac{x_{vmax} - x_{vmin}}{x_{wmax} - x_{wmin}} \\ &= \frac{90 - 6}{35 - (-5)} \\ &= 2.1 \end{aligned}$$

$$\begin{aligned} S_y &= \frac{y_{vmax} - y_{vmin}}{y_{wmax} - y_{wmin}} \\ &= \frac{80 - 17}{40 - 10} \\ &= 2.1 \end{aligned}$$

$$C_m = T(x_{vmin}, y_{vmin}) \cdot S_{S_x, S_y} \cdot T(-x_{wmin}, -y_{wmin})$$

$$\begin{aligned} &= \begin{pmatrix} 1 & 0 & 6 \\ 0 & 1 & 17 \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} 2.1 & 0 & 0 \\ 0 & 2.1 & 0 \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} 1 & 0 & 5 \\ 0 & 1 & -10 \\ 0 & 0 & 1 \end{pmatrix} \\ &= \begin{pmatrix} 2.1 & 0 & 16.5 \\ 0 & 2.1 & -4 \\ 0 & 0 & 1 \end{pmatrix} \end{aligned}$$

The points mapped to viewport will be A' and B'.

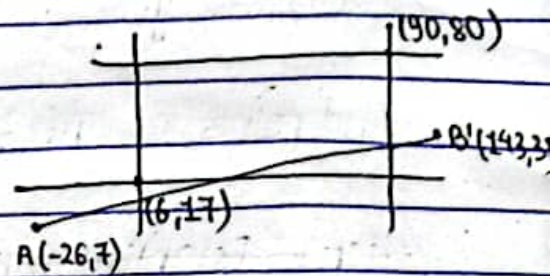
$$P' = C_m \cdot P$$

$$\begin{aligned} &= \begin{pmatrix} 2.1 & 0 & 16.5 \\ 0 & 2.1 & -4 \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} -20 & 60 \\ 5 & 20 \\ 1 & 1 \end{pmatrix} = \begin{pmatrix} -25.5 & 142.5 \\ 6.5 & 38 \\ 1 & 1 \end{pmatrix} \end{aligned}$$



so,  $A'(-25.5, 38)$  and  $B'(142.5, 38)$   
 $\approx A'(-26, 7)$  and  $B'(143, 38)$

Now, Let's find intersection points.



Step 1: Develop region codes for line end points

| $A'(-26, 7)$  |                                            |   | $B'(143, 38)$ |                                            |   |
|---------------|--------------------------------------------|---|---------------|--------------------------------------------|---|
| $L: -26 < 6$  | T                                          | 1 | $L: 143 < 6$  | F                                          | 0 |
| $R: -26 > 90$ | F                                          | 0 | $R: 143 > 90$ | T                                          | 1 |
| $B: 7 < 17$   | T                                          | 1 | $B: 38 < 17$  | F                                          | 0 |
| $T: 7 > 80$   | F                                          | 0 | $T: 38 > 80$  | F                                          | 0 |
| Region        | $(0101)$<br><del><math>(1010)</math></del> |   | Region        | $(0010)$<br><del><math>(0100)</math></del> |   |

So, this is not the case of full visibility.

Step 2: Perform AND operation of region code of end points.

$$\begin{array}{r} 1010 \\ 0100 \\ \hline 0000 \end{array}$$

So, this is the case of partial visibility.

Step 3: Find intersection points for end points.

For  $B'$  slope  $(m) = \frac{y_2 - y_1}{x_2 - x_1} = \frac{38 - 7}{143 - 26} = \frac{31}{117}$

For  $B'(143, 38)$ ,

$$x = x_{\max} = 90$$

$$y = m(x - x_1) + y_1$$

$$= \frac{31}{117}(90 - 143) + 38$$

$$\therefore y = 28.27 \approx 28 \quad (90, 28)$$



For  $A'(-26, 7)$ ,

For bit 1,

$$x = x_{\min} = 6$$

$$y = m(x - x_1) + y_1 \\ = \frac{31}{169} (6 + 26) + 7$$

$$\therefore y = 12.86$$

$$\text{So, } (x', y') = (6, 12.86)$$

For bit 3,

$$y = y_{\min} = 17$$

$$x = \frac{1}{m} (y - y_1) + x_1 \\ = \frac{169}{31} (17 - 12.86) + 6$$

$$\therefore x = 28.57 \approx 29$$

$$\text{So, } (29, 17)$$

Hence, final coordinates of visible line end points are

$$\underline{\underline{A(29, 17) \text{ and } B(90, 28)}}$$

Q16) Reduce a triangle with coordinates  $A(0,0)$ ,  $B(0,4)$ ,  $C(8,4)$ ,  $D(8,0)$  to twice its original about a fixed point  $P(4,2)$  also derive a transformation matrix to convert this rectangle into a square.

soln

$$C_m = T_{(4,2)} \cdot S_{s_x=2, s_y=2} \cdot T_{(-4,-2)} \\ = \begin{pmatrix} 1 & 0 & 4 \\ 0 & 1 & 2 \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} \frac{1}{2} & 0 & 0 \\ 0 & \frac{1}{2} & 0 \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} 1 & 0 & -4 \\ 0 & 1 & -2 \\ 0 & 0 & 1 \end{pmatrix}$$



$$\therefore C_m = \begin{pmatrix} 0.5 & 0 & -0.2 \\ 0 & 0.5 & -0.1 \\ 0 & 0 & 1 \end{pmatrix}$$

$$P' = C_m \cdot P$$

$$= \begin{pmatrix} 0.5 & 0 & -0.2 \\ 0 & 0.5 & -0.1 \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} 0 & 0 & 8 & 8 \\ 0 & 4 & 4 & 0 \\ 1 & 1 & 1 & 1 \end{pmatrix}$$

$$= \begin{pmatrix} -4 & 4 & 12 & 12 \\ -2 & 6 & 6 & -2 \\ 1 & 1 & 1 & 1 \end{pmatrix} = \begin{pmatrix} 2 & 2 & 6 & 6 \\ 1 & 3 & 3 & 1 \\ 1 & 1 & 1 & 1 \end{pmatrix}$$

So, So, rectangle coordinates after reduction are  
 $(2, 1), (2, 3), (6, 3), (6, 1)$  Ans

To convert rectangle to square, Lets calculate 'l' & 'b' for rectangle,

$$l = 8 - 0 = 8$$

$$b = 4 - 0 = 4$$

So, Lets shorten the length from 8 to 4 in x-axis to be square.

$$\text{So, } S_x = \frac{4}{8} = 0.5$$

$$S_y = 1$$

So, Using fixed point, we first translate the fixed point to center (0,0),  
 scale by  $S_x = 0.5, S_y = 1$  and translate back.

The transformation matrix will be,

$$C_m = T(x_f, y_f) \cdot S_{S_x=0.5, S_y=1} \cdot T(-x_f, -y_f)$$



$$= \begin{pmatrix} 1 & 0 & 4 \\ 0 & 1 & 2 \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} 0.5 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} 1 & 0 & -4 \\ 0 & 1 & -2 \\ 0 & 0 & 1 \end{pmatrix}$$

$$= \begin{pmatrix} 0.5 & 0 & 2 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} \underline{\underline{Ans}}$$

Q17) A triangle with vertices A(0,10), B(10,0), C(-10,0) is required to be shifted down by 5 units, then rotate in anticlockwise direction by  $30^\circ$  and scale by twice its original size. What will be the final location?

Soln

Step 1: Shift down by 5 units i.e.  $T_{(-5, -5)}$

Step 2: Rotate ccw by  $30^\circ$  i.e.  $R_{\theta_{ccw}^{30^\circ}}$

Step 3: Scale by  $S_x = 2, S_y = 2$

$$SO, C_m = S_{S_x=2, S_y=2} \cdot R_{\theta_{ccw}^{30^\circ}} \cdot T_{(-5, -5)}$$

$$= \begin{pmatrix} 2 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} \cos 30^\circ & -\sin 30^\circ & 0 \\ \sin 30^\circ & \cos 30^\circ & 0 \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} 1 & 0 & -5 \\ 0 & 1 & -5 \\ 0 & 0 & 1 \end{pmatrix}$$

$$= \begin{pmatrix} 1.73 & -1 & -3.66 \\ 1 & 1.73 & -13.66 \\ 0 & 0 & 1 \end{pmatrix}$$

$$SO, P' = C_m \cdot P$$

$$= \begin{pmatrix} 1.73 & -1 & -3.66 \\ 1 & 1.73 & -13.66 \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} 0 & 10 & -10 \\ 10 & 0 & 0 \\ 1 & 1 & 1 \end{pmatrix}$$



$$= \begin{pmatrix} -13.66 & 13.66 & -20.98 \\ 3.66 & -3.66 & -23.66 \\ 1 & 1 & 1 \end{pmatrix}$$

$$\approx \begin{pmatrix} 14 & 14 & -21 \\ 4 & -4 & -24 \\ 1 & 1 & 1 \end{pmatrix}$$

Hence, final location will be A(14, 4), B(14, -4), C(-21, -24). Ans

Q18) Rotate triangle A(0,0), B(1,1), C(5,2) about origin and about point P(-1,-1) by  $45^\circ$  in a cc direction.

Soln

$$\text{Composite matrix } (C_m) = T_{(-1,-1)} \cdot R_{0_{ccw}^{45^\circ}} \cdot T_{(1,1)} \cdot R_{0_{ccw}^{45^\circ}}$$

$$= \begin{pmatrix} 1 & 0 & -1 \\ 0 & 1 & -1 \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} \cos 45^\circ & -\sin 45^\circ & 0 \\ \sin 45^\circ & \cos 45^\circ & 0 \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} 1 & 0 & 1 \\ 0 & 1 & 1 \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} \cos 45^\circ & -\sin 45^\circ & 0 \\ \sin 45^\circ & \cos 45^\circ & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

$$= \begin{pmatrix} 0 & -1 & -1 \\ 1 & 0 & 0.414 \\ 0 & 0 & 1 \end{pmatrix}$$

$$P' = C_m \cdot P$$

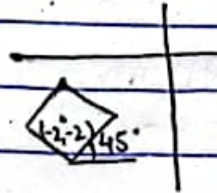
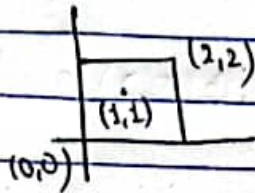
$$= \begin{pmatrix} 0 & -1 & -1 \\ 1 & 0 & 0.414 \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} 0 & 1 & 5 \\ 0 & 1 & 2 \\ 1 & 1 & 1 \end{pmatrix}$$

$$= \begin{pmatrix} -1 & -2 & -3 \\ 0.414 & 1.414 & 5.414 \\ 1 & 1 & 1 \end{pmatrix}$$

Final points: A'(-1, 0), B'(-2, 1), C'(-3, 5) Ans



Q19) Write the series of transformation in matrix that are needed to place the square shown in the figure A reduced to one third of its original size, into the position shown in fig. B where the center of square is at  $(-2, -2)$ .



soln

$$C_m = T_{(-2, -2)} \cdot R_{45^\circ \text{ ccw}} \cdot S_{\frac{1}{3}, \frac{1}{3}} \cdot T_{(1, 1)}$$

$$= \begin{pmatrix} 1 & 0 & -2 \\ 0 & 1 & -2 \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} \cos 45^\circ & -\sin 45^\circ & 0 \\ \sin 45^\circ & \cos 45^\circ & 0 \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} \frac{1}{3} & 0 & 0 \\ 0 & \frac{1}{3} & 0 \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} 1 & 0 & -1 \\ 0 & 1 & -1 \\ 0 & 0 & 1 \end{pmatrix}$$

$$= \begin{pmatrix} 0.236 & -0.236 & -2 \\ 0.236 & 0.236 & -2.47 \\ 0 & 0 & 1 \end{pmatrix}$$

To prove,  $P' = C_m \cdot P$

$$= \begin{pmatrix} 0.236 & -0.236 & -2 \\ 0.236 & 0.236 & -2.47 \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix}$$

$$= \begin{pmatrix} -2 \\ -2 \\ 1 \end{pmatrix} //$$

Hence, series is as :

Step 1: Translation by factor  $(-1, -1)$  to bring object to origin.  
(fixed point)

Step 2: scaling by factor  $\frac{1}{3}$  in both axis.



Step 3: Rotation by  $45^\circ$  in counter clockwise direction.

Step 4: Translate ~~to~~ by factor  $(-2, -2)$  to move fixed point to req<sup>d</sup> point as in fig.

Q20) Reflect a ~~point~~ Triangle with vertices  $A(4, 1)$ ,  $B(5, 2)$  and  $C(4, 3)$  about line  $3x - 4y + 8 = 0$ .

Soln

Given line,  $-4y = -3x - 8$

or,  $y = \frac{3}{4}x + 2$

Here,  $m = \frac{3}{4}$  ;  $c = 2$

so,  $\theta = \tan^{-1}\left(\frac{3}{4}\right) = 36.87^\circ$

For reflection,

$$C_m = T_{(0, -2)} \cdot R_{\theta}^{36.87^\circ} \cdot R_{fx} \cdot R_{\theta}^{-36.87^\circ} \cdot T_{(0, -2)}$$

$$= \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 2 \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} \cos 36.87^\circ & -\sin 36.87^\circ & 0 \\ \sin 36.87^\circ & \cos 36.87^\circ & 0 \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} 1 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} \cos 36.87^\circ & \sin 36.87^\circ & 0 \\ -\sin 36.87^\circ & \cos 36.87^\circ & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

$$\begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & -2 \\ 0 & 0 & 1 \end{pmatrix}$$

$$= \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 2 \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} 0.279 & 0.960 & -1.92 \\ 0.960 & -0.279 & 0.559 \\ 0 & 0 & 1 \end{pmatrix}$$

$$= \begin{pmatrix} 0.279 & 0.960 & -1.92 \\ 0.960 & -0.279 & 2.559 \\ 0 & 0 & 1 \end{pmatrix}$$



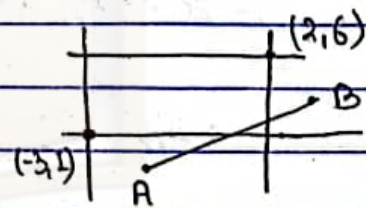
$$\begin{aligned}
 S_0, P' &= C_m \cdot P \\
 &= \begin{pmatrix} 0.279 & 0.960 & -1.92 \\ 0.960 & -0.279 & 2.559 \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} 4 & 5 & 4 \\ 1 & 2 & 3 \\ 1 & 1 & 1 \end{pmatrix} \\
 &= \begin{pmatrix} 0.159 & 1.39 & 2.079 \\ 6.12 & 6.80 & 5.560 \\ 1 & 1 & 1 \end{pmatrix}
 \end{aligned}$$

Hence, Final triangle vertices are approximately,  
 $A(0,6), B(1,7), C(2,6)$  Ans

Q21) Clip a line with end points A(1,-2) and B(3,3) against a clip window with its lower left corner at (-3,1) and upper right corner at (2,6).

Soln

Line end points: A(1,-2), B(3,3)



Step 1: Develop region codes for line end points.

A(1,-2):

L:  $1 < -3$  F 0

R:  $1 > 2$  F 0

B:  $-2 < 1$  T 1

T:  $-2 > 6$  F 0

Code: (0100)

B(3,3)

L:  $3 < -3$  F 0

R:  $3 > 2$  T 1

B:  $3 < 1$  F 0

T:  $3 > 6$  F 0

Code: (0010)

So, this is not the case of full visibility.



Step 2: Perform AND operation of region codes of end points.

$$\begin{array}{r} 0100 \\ 0010 \\ \hline 0000 \end{array}$$

So, this is the case of: partial visibility.

Step 3: Find intersection points with clip window.

For A, (1, -2)

$$m = \frac{y_2 - y_1}{x_2 - x_1} = \frac{3 - (-2)}{3 - 1} = \frac{5}{2}$$

$$y = y_{\min} = 1$$

$$x = \frac{1}{m}(y - y_1) + x_1$$

$$= \frac{2}{5}(1 + 2) + 1$$

$$= 2.2$$

$$(2.2, 1) \approx (2, 1)$$

For B, (3, 3)

$$x = x_{\min} = 2$$

$$y = m(x - x_1) + y_1$$

$$= \frac{5}{2}(2 - 3) + 3$$

$$= 0.5$$

$$(2, 0.5) \approx (2, 1)$$

Hence, the visible end points of line are A(2, 1) and B(2, 1)

i.e. line passes through intersection point of clip window.



Q22) Prove that reflection about line  $y = -x$  is equivalent to reflection about  $y$ -axis followed by a counter clockwise rotation by  $90^\circ$ .

Ans

Case I : Reflection about line  $y = -x$ .

Here,  $m = -1$

$$\text{or, } \theta = \tan^{-1}(-1)$$

$$\therefore \theta = -45^\circ$$

$$C_m = R_{\theta_{ccw}}^{-45^\circ} \cdot R_{fy} \cdot R_{\theta_{ccw}}^{+45^\circ}$$

$$= \begin{pmatrix} \cos 45^\circ & \sin 45^\circ & 0 \\ -\sin 45^\circ & \cos 45^\circ & 0 \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} 1 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} \cos 45^\circ & -\sin 45^\circ & 0 \\ \sin 45^\circ & \cos 45^\circ & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

$$\therefore C_{m1} = \begin{pmatrix} 0 & -1 & 0 \\ -1 & 0 & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

Case II : counter clockwise rotation by  $90^\circ$  after  $R_{fy}$ .

$$C_m = R_{\theta_{ccw}}^{90^\circ} \cdot R_{fy}$$

$$= \begin{pmatrix} \cos 90^\circ & -\sin 90^\circ & 0 \\ \sin 90^\circ & \cos 90^\circ & 0 \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} -1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

$$\therefore C_{m2} = \begin{pmatrix} 0 & -1 & 0 \\ -1 & 0 & 0 \\ 0 & 0 & 1 \end{pmatrix}$$



As the composite matrix obtained in both cases are same, reflection about  $y = -x$  is equivalent to reflection about  $y$ -axis followed by a counter clockwise rotation by 90 degrees.